



Securing understanding of ratio as a multiplicative relationship

- 1 A ratio shows the relative sizes of 2 or more values and allows you to compare a part with another part in a whole. The relationship between parts and/or the whole is always _____. Tick 1 correct answer

- ☐ additive
- ☒ multiplicative
- ☐ constant
- ☐ subtractive

- 2 A robot is travelling around a toxic planet. The protective clothing is heavy, so for every 6 protective items, there must be 2 batteries. Match the equipment list to its effect on the robot. Write the correct letter in each box

a	18 protective and 6 batteries
b	3 protective and 2 batteries
c	12 protective and 2 batteries

c	This means the robot is protected but slow.
a	This means the robot is fully protected and quick.
b	This means the robot is quick but not fully protected.

- 3 To make a "Sunrise" orange paint, 2 cans of yellow paint for every 1 can of red paint is used. Match the paint shades to the recipes. Write the correct letter in each box

a	"Sunrise" Orange
b	Darker than "Sunrise" orange
c	Lighter than "Sunrise" orange

b	6 cans of yellow and 4 cans of red
c	10 cans of yellow and 4 cans of red
a	4 cans of yellow and 2 cans of red

- 4 To make lemonade, 1 lemon and 2 tablespoons of sugar are used for every 500 ml of water. Jun uses 750 ml of water to make lemonade. Jun should use 3 tablespoons of sugar. Fill in the blank



5 Select all the statements that are correct. Tick 2 correct answers

- ☒ A ratio describes the relationship between values.
- ☐ Ratio uses an additive relationship.
- ☐ Adding the same amount to each part in a ratio keeps the ratio the same.
- ☒ Multiplying the same amount to each part in a ratio keeps the ratio the same.

6 "Choko Milk chocolate" bars are made with 2 parts cocoa, 5 parts milk powder and 1 part sugar. Too much cocoa makes it too bitter, too much sugar makes it too sweet. Match each result with its recipe Write the correct letter in each box

a	Too sweet
b	Too bitter
c	Perfect
d	Too much milk powder

<i>a</i>	18 parts cocoa, 45 parts milk powder and 10 parts sugar.
<i>b</i>	15 parts cocoa, 35 parts milk powder and 7 parts sugar.
<i>c</i>	10 parts cocoa, 25 parts milk powder and 5 parts sugar.
<i>d</i>	20 parts cocoa, 55 parts milk powder and 10 parts sugar.

